Abstract – SMDM deadline 15 January 2025; 375 word maximum

Title: Associations and trends between county-level social vulnerability and drug overdose deaths in the United States

Purpose

To investigate the association of county-level Social Vulnerability Index (SVI) changes across time on drug overdose death rates in the United States (US) from 2014 to 2020.

Methods

We conducted a serial cross-sectional study to evaluate the association between county-level social vulnerability with drug overdose deaths in the US from 2014 to 2020. Our primary aim was to evaluate the associations between SVI and its four themes (Socioeconomic Status, Household Characteristics, Racial & Ethnic Minority Status, and Housing Type & Transportation) with drug overdose deaths by year (2014, 2016, 2018, and 2020). Our secondary aim evaluated the interaction between SVI and its four themes with time on drug overdose deaths.

Results

There were totals of 3137, 3135, 3141, and 3141 counties with the SVI and drug overdose death rate data for 2014, 2016, 2018, and 2020, respectively. Drug-related overdose death rates increased within each SVI and its thematic quintile across the years. In the state-adjusted models, increasing county-level SVI quintile was statistically significantly associated with increased drug overdose deaths rate across all years studied. In 2014, going from an SVI from 0 to 1 was associated with an increase of 4.74 drug overdose deaths per 100,000 persons (95% CI: 3.45, 6.04). In 2016, going from an SVI from 0 to 1 was associated with an increase of 2.83 drug overdose deaths per 100,000 persons (95% CI: 1.54, 4.13). In 2018, going from an SVI from 0 to 1 was associated with an increase of 1.74 drug overdose deaths per 100,000 persons (95% CI: 0.44, 3.03). In 2020, going from an SVI from 0 to 1 was associated with an increase of 1.94 drug overdose deaths per 100,000 persons (95% CI: 0.66, 3.22).

Conclusions

Overall, counties with high levels of social vulnerability had higher rates of drug overdose deaths between 2014 and 2020. Although drug overdose death rates increased during this period, the incremental increase in drug overdose death rates across time was lower in counties that were highly vulnerable compared to counties that were least vulnerable. Additionally, the strengths of the association decreased from 2014 to 2018 but increased in 2020. These findings may inform policymakers as they strategize plans to incorporate county-level social vulnerability to reduce drug overdose deaths.